Fiber Optics - Cut-off frequencies and spectral attenuation report

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I. Introduction

In this experiment we will measure two fundamental characteristics of optical fibers: the cut-off frequencies of higher order modes, and the spectral attenuation, focusing on the wavelength interval [650, 1750] nm. It is well known, from the modal theory, that optical fibers support a frequency-dependent count of propagation modes, and that each mode, with the exception of the fundamental mode, is allowed to propagate only if the wavelength of the field is shorter than a cut-off wavelength, which is determined by the geometry of the refractive index profile of the fiber. In a general case, the so-called dispersion diagram is a key tool in understanding this behaviour: in Fig. 1 the (normalized) propagation constant of each mode is given for each (normalized) frequency, and it is clear that for all the modes except for the fundamental $LP_{0,1}$, there is a minimum frequency that allows propagation.

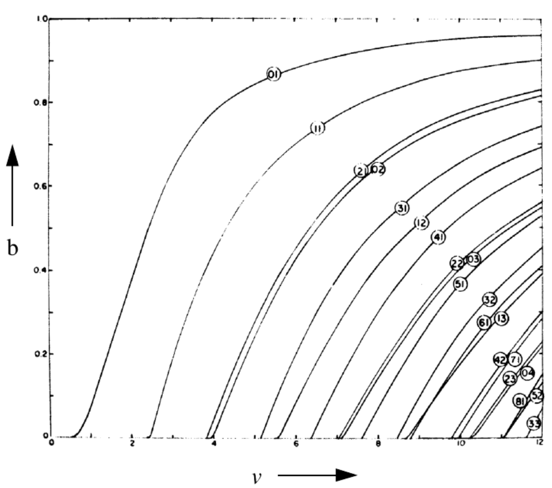

Fig. 1: Propagation constant vs. normalized frequency (note the cut-off frequencies).

The characterization of the cut-off points is done by measuring the optical power at the various wavelengths: an important result of modal theory is that the modes are orthogonal, so the overall power transmitted along the fiber is just the sum of the power carried by each mode (all the cross-terms are 0 in the expansion for $|\mathbf{E}|^2$). The cut-off frequency would then be determined by spotting the point in which the total power changes, which indicates that a particular mode is allowed to propagate. However, a simple measurement of the power spectrum is not sufficient because it lacks the account for the spectral attenuation of the fiber and we don't have a way to measure the injected power per mode at the input. So another strategy is actually used: if we introduce a *bend* in the fiber sample, we expect the cut-off frequency to shift toward shorter wavelengths.

By computing the difference between the transmission spectra of the bent and straight fiber we observe the presence of peaks near the actual spot of the cut-off wavelength.

II. Experimental setup

The test setup for the cut-off frequency assessment is straightforward: a broadband light source is injecting light into the fiber under test (we suppose that all the modes we are observing are excited by the source), and at the other side an optical spectrum analyzer carries out the transmitted power measurement. The fiber samples we are testing are $\approx$ 2m long, and are of the following types:

1) Non-zero dispersion-shifted fiber (NZDSF).
2) Low bending loss fiber.
3) Simple step-index fiber.

The curvature radius of the bend is not actually critical, but to highlight certain characteristics of the response, after measuring a first response, we decided to apply another bend of greater or smaller radius.

In the second part, after assessing the reference response of the setup cables, we measured the power spectrum transmitted by two fiber bobbins, the first one being a standard fiber and the second one being a DCF designed to compensate for -101 ps/nm.

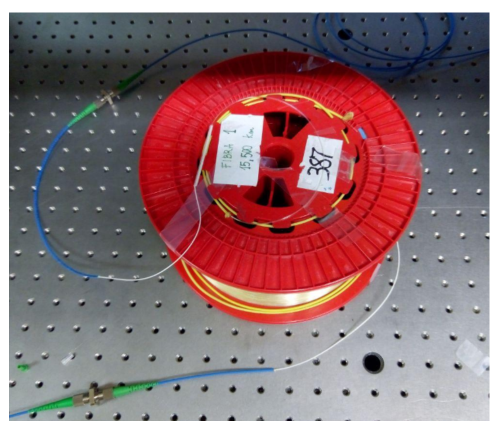

Fig. 2: Fiber under test and connection to source and analyzer.

III. Results

A. Measure of the cut-off frequencies

In the first part of the experience, three sets of data were recorded for each of the three fibers under testing. The first set of measurements is used as reference of the overall system, and was taken by applying no bending to the fiber in question.

On the other hand, the other two measurements were obtained by taping the bent fiber to the optical table, where each measure features a different degree of bend to it. All the measurements were obtained by the output of the optical spectrum analyzer (OSA), hence they represent the received power as a function of wavelength. An example of the three measures is shown in Figure 3.

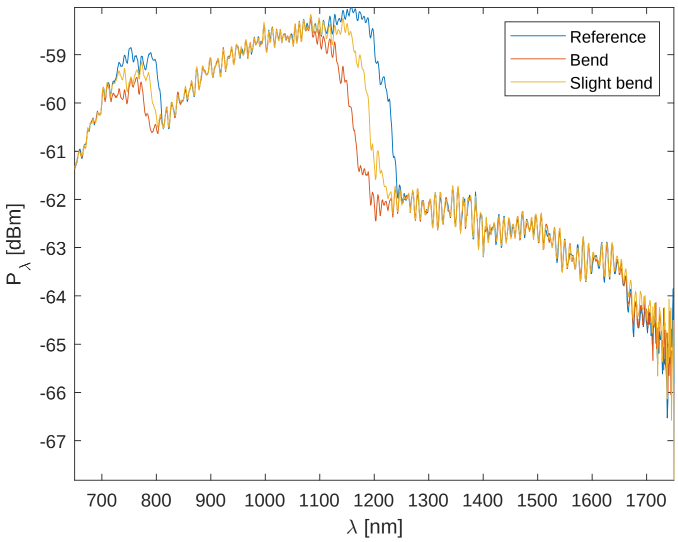

Fig. 3: Comparison of the three output measures.

Even before performing the real analysis, one can notice the steep decrease in power after two peaks at about 850 and 1200 nm, which is the typical behavior of the spectrum when a cut-off wavelength is met. Moreover, the spectrum is evidently more attenuated as the fiber is bent, since the slightly-bent fiber measurement is always placed between the bent and the reference fiber (except for the final portion of the graph, where noise is too strong to draw any conclusions from that).

We'll begin our analysis with the non-zero dispersion-shifted fiber. The spectral bending attenuation losses were calculated by subtracting the reference measure to the measures with bend. The results are shown in Figure 4, where the wavelength axis was slightly cut in order to hide the irrelevant noise after 1600 nm.

From this portion of the spectrum, two main alterations are distinguishable. The first one from the right is clearly the $\mathrm{LP}_{1,1}$ mode, set at about 1200 nm. The attenuation caused by the tighter bend is quite large, going beyond 3 dB (namely a 50% power reduction) at the maximum point; the slightly-bent fiber has a considerable attenuation, as well. Another interesting thing can be inferred from this peak, namely the presence of tinier peaks on the overall shape. These local maxima can be explained in terms of degeneracy (or rather, non-degeneracy) of modes. Since it is not an ideal fiber, the four-fold degeneracy of $\mathrm{LP}_{n,p}$ modes (with $n \geq 1$) is slightly altered, causing the spectra of the modes to marginally shift in different ways.

The second portion, between 700 and 800 nm, shows an even clearer separation between two peaks. As a matter of fact,

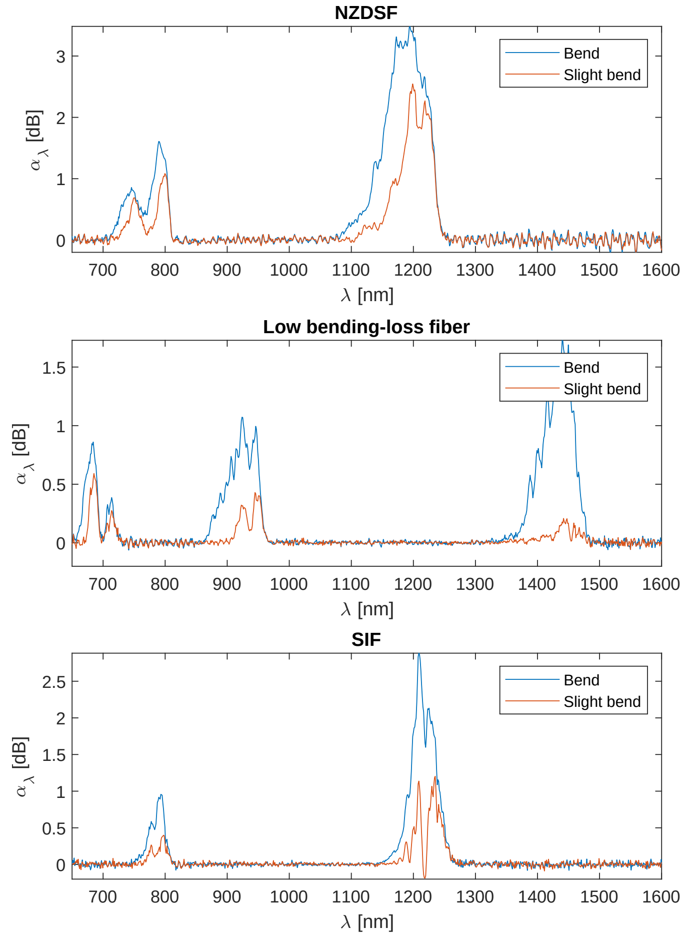

Fig. 4: Spectral attenuation of the three fibers.

these are actually two different modes, the $\mathrm{LP}_{2,1}$ on the right and the $\mathrm{LP}_{0,2}$ on the left, which are very near each other, as also shown in Figure 1. In this case there is no clear distinction between quasi-degenerate modes, but since the $\mathrm{LP}_{0,2}$ has $n = 0$, its degeneracy is only two-fold, and it is given by the axial symmetry of the fiber, which make the spatial distribution of $\mathrm{LP}^{(x)}$ and $\mathrm{LP}^{(y)}$ modes identical (supposing absence of PMD).

The second fiber is optimized to have low bending losses, and this is evident from the decrease in the maximum attenuation (which peaks at 1.72 dB for the first cut-off frequency). The fiber shows an even lower response when a less severe bend is applied, barely registering a peak at about 0.19 dB for the first cut-off. As the wavelength decreases and other higher-order modes appear, the mitigation of attenuation seems lower. This may indicate that the fiber is optimized for bending losses that occur at lower frequencies, and that this optimization becomes less effective as the frequency is increased.

Similar considerations to the first fiber can be also applied to this second one. Moreover, two additional things can be observed. First, the cut-off wavelengths are considerably shifted to the right with respect to the first fiber. Second, because of this red-shift two additional peaks appear within the measured range. By looking at the normalized dispersion relation graph, they possibly correspond to the $\mathrm{LP}_{3,1}$ modes (which are non degenerate, considering the two peaks). Overall, the estimated cut-off wavelengths are 1450, 950, 930, 715 and 685 nm.

Finally, the traditional step-index fiber is considered. The cut-off wavelengths of this fiber are a little shifted with respect to the first fiber, at approximately 1220 nm, 800 nm and 780 nm. The overall attenuation lies between the first and the second fiber. In this fiber, the quasi-degenerate modes for the $LP_{1,1}$ mode look particularly separated, especially for the measure of the slightly-bent fiber. This could be caused by a larger amount of polarization mode dispersion, but the curve is not well-defined enough to be sure.

B. Measure of the attenuation spectrum

The second part deals with the measurement of the attenuation spectrum of two fiber bobbins.

Similarly to the first part, a reference measurement was taken by connecting directly the broadband source with the OSA. Again, the spectra from the two bobbins were thus subtracted by this reference measure, in order to remove the attenuation given by the other setup components. The final spectra are shown in Figure 5.

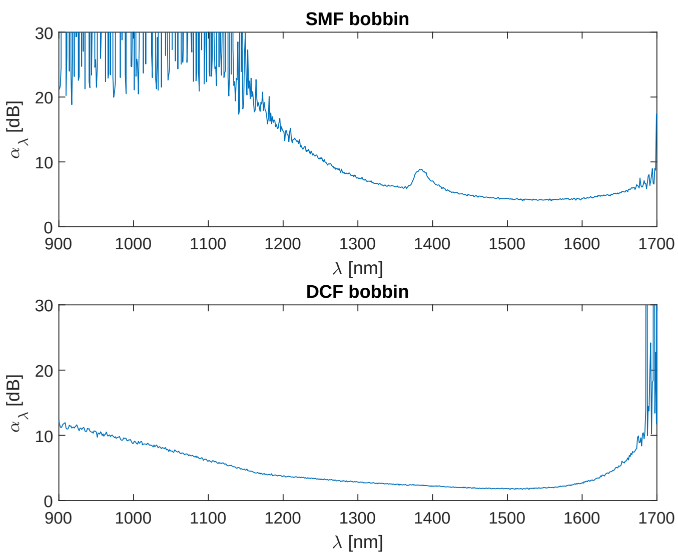

Fig. 5: Attenuation spectrum of the two fiber bobbins.

The first bobbin is a single-mode fiber bobbin. Its spectrum is shown between 900 and 1700 nm because, even in this case, noise overtakes the signal for the other portions of the spectrum; actually, the effect of noise is already visible below 1200 nm.

The minimum is found, as expected, in the third window, more precisely at 1520 nm. Given the value of attenuation $\alpha = 4.04$ dB and the fiber length $L_1 = 15.50$ km, the attenuation coefficient is 0.26 dB/km. Although typical values of modern fibers are a little lower, at about 0.2 dB/km, this fiber bobbin still shows good performance for today's standard.

Going towards the shorter wavelengths, Rayleigh scattering increases with the fourth power of the frequency, and is the main contributor of attenuation. On the other hand, for wavelengths longer than those of the third window, the main attenuation effect is given by molecular vibrations. Since this is a single-mode fiber, one may also infer that the small bump at 1380 nm is caused by one of the resonant frequencies of hydroxide ions OH^-, a typical contaminant that decreases performance in optical fibers. The other resonant frequency in this range, which is found at 1240 nm, shows no increase in attenuation, possibly due to the increasing noise at the end of the graph, but also because this peak is much less pronounced than that at 1380 nm.

The second bobbin is dispersion-compensating. The length of this fiber is unknown, but it can be estimated by making some consideration about the minimum attenuation value. The minimum attenuation $\alpha = 1.7$ dB is found at 1530 nm. Assuming standard performance typical of modern fibers (as previously states, about 0.2 dB/km), the length of this fiber should not be longer than 8.5 km.

The overall attenuation performance of this second fiber seems even greater than standard fibers. There are no local maxima that suggest resonant absorption by hydroxide ions, and the effect of Rayleigh scattering seems very restrained at lower wavelengths when compared to the first single-mode fiber. One potential explanation of this behavior is that the bobbin is actually much shorter than the estimated length, and that the attenuation at 1530 nm is moderately larger than 0.2 dB/km. This would explain the particularly flat absorption spectrum.

IV. Conclusions

This experiment was divided into two main parts: at first, we evaluated the cut-off frequencies for different types of fibers, while in the second part, we reported the attenuation spectrum of two fiber bobbins.

In the introduction (Section I), we briefly explained the concept of cut-off frequencies. Then, by taking advance of the orthogonality between modes, we showed the relation between the changes in optical power and the cut-off frequencies.

In Section III-A, we measured the cut-off frequency of three different types of fibers. Interesting results are the red-shift of the cut-off frequencies for the low bending-loss fiber and its attenuation behaviour: in fact the attenuation gradually increase while moving to lower wavelengths. Also, we noticed the presence of non-degenerate modes near the attenuation peaks (Fig.4), in particular for the lower modes, a phenomenon present in any real fiber to a certain degree.

Finally, in Section III-B we drew the attenuation spectrum of two different fiber spools. We ended up with some expected results, such as the attenuation minimum in the third window, the presence of strong absorption peaks due to the resonant frequencies of hydroxide-ion impurities in the fiber, and the lower-wavelength absorption enhancement caused by Rayleigh scattering.

However, we also reported unexpected results, as the flattened attenuation spectrum from the DCF bobbin. This behaviour could be justified by supposing that fiber was fabricated through techniques, such as VAD or OVD, that were able to drastically reduce contamination from OH^- ions.

In conclusion, we showed and commented on the simultaneous presence of several propagating modes; moreover, we discussed some of the main causes for absorption in optical fibers.